THE COORDINATION CONTAGION

How Trust and Cooperation Spread Through International Networks: An Epidemiological Model of Civilizational Coordination

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Abstract

Coordination capacity does not develop in isolation. This paper introduces an epidemiological framework for understanding how trust, cooperation, and institutional effectiveness spread through international networks. Using the K-Index as our measure of coordination capacity, we model its diffusion across 193 nations over 74 years (1950-2024) through trade networks, treaty alliances, communication flows, and geographic proximity. We find that coordination exhibits strong contagion dynamics: a 0.1-point increase in a country's trading partners' average K-Index predicts a 0.03-0.05 point increase in that country's K-Index within 5 years ($p < 0.001$). We identify "super-spreader" nations (particularly Nordic countries and successful Asian tigers) whose coordination improvements disproportionately affect global K trajectories. The model reveals a "coordination herd immunity" threshold: when approximately 65% of nations by GDP-weighted influence exceed K = 0.65, coordination collapse becomes self-limiting in connected networks. These findings suggest that strategic coordination investments in high-connectivity nodes could accelerate global cooperation capacity far beyond linear projections.

Keywords: coordination contagion, network diffusion, trust spillovers, institutional spread, epidemiological models, K-Index

Contents

1 Introduction

Why did democratization spread through Eastern Europe in 1989 like a cascade rather than proceeding incrementally? Why do economic crises propagate across borders even between countries with limited direct trade? Why did institutional reforms in one East Asian nation seem to catalyze similar reforms in neighbors within years?

The answer, we propose, is that **coordination is contagious**. Trust begets trust. Successful cooperation in one context creates templates, expectations, and infrastructure that lower the costs of cooperation elsewhere. Institutional innovations diffuse through networks of observation, imitation, and direct transmission.

This paper develops a formal epidemiological model of coordination spread, treating the K-Index—a comprehensive measure of societal coordination capacity (Stoltz, 2025)—as the "pathogen" whose diffusion we track. Unlike actual pathogens, higher coordination is beneficial, but the mathematical structure of spread is remarkably similar: exposure through network ties, probability of "infection" (adoption), incubation periods, and recovery/decay dynamics.

1.1 The Puzzle of Coordination Clustering

A striking feature of global coordination capacity is its geographic and network clustering. High-K nations cluster together: the Nordic countries, Western Europe, and the Anglosphere form one cluster; East Asian tigers form another. Low-K nations similarly cluster: the Sahel, Central Africa, and parts of Central Asia.

This clustering could arise from three mechanisms:

1. **Common Causes**: Shared history, geography, or culture produce similar coordination outcomes independently
2. **Selection Effects**: Nations with similar coordination capacities choose to interact more
3. **Contagion**: Coordination capacity actually spreads through network ties

Distinguishing these mechanisms has profound policy implications. If clustering is purely due to common causes, targeted interventions in individual nations may succeed. If primarily selection, changing interaction patterns matters more than domestic reform. If contagion dominates, strategic investments in high-connectivity nodes could produce cascading benefits far beyond the direct targets.

1.2 Research Questions

This paper addresses four core questions:

1. **RQ1**: Does coordination capacity spread through international networks after controlling for common causes and selection?

2. **RQ2**: What network channels—trade, treaties, communication, proximity—transmit coordination most effectively?

3. **RQ3**: Are there "super-spreader" nations whose coordination improvements disproportionately affect global trajectories?

4. **RQ4**: Does a "coordination herd immunity" threshold exist beyond which collapse becomes self-limiting?

2 Theoretical Framework

2.1 The Epidemiology of Social Phenomena

The application of epidemiological models to social phenomena has a distinguished history. Bass (1969) pioneered diffusion models for product adoption. Christakis & Fowler (2007) demonstrated contagion in obesity and smoking through social networks. Centola (2018) formalized conditions under which "complex contagions"—behaviors requiring multiple exposures—spread through networks.

Coordination capacity presents a particularly interesting case for contagion modeling because it is inherently *relational*. Unlike individual behaviors (smoking, product adoption), coordination capacity is partially constituted by relationships with others. A nation's ability to cooperate depends partly on whether its partners are trustworthy and institutionally competent.

2.2 The K-Index Contagion Model

Let $K_i(t)$ denote country i's K-Index at time t. We model its dynamics as:

$$\frac{dK_i}{dt} = \underbrace{\beta \sum_j A_{ij}(t)K_j(t)(K_{max} - K_i(t))}_{\text{Contagion from network}} - \underbrace{\gamma K_i(t)}_{\text{Decay}} + \underbrace{\eta_i(t)}_{\text{Exogenous shocks}} + \underbrace{\mu_i}_{\text{Baseline growth}} \tag{1}$$

Where:

- $A_{ij}(t)$ is the network adjacency matrix (normalized weights of ties between i and j)
- β is the contagion rate (how strongly network exposure influences K)
- K_{max} is the theoretical ceiling (we use 1.0)

- γ is the decay rate (how quickly coordination erodes without reinforcement)
- $\eta_i(t)$ captures exogenous shocks (wars, revolutions, natural disasters)
- μ_i is country-specific baseline growth (capturing path dependence)

The key term is the contagion expression: $\beta \sum_j A_{ij} K_j (K_{max} - K_i)$. This captures:

1. Exposure depends on network ties (A_{ij}) and partner quality (K_j)
2. Adoption probability decreases as K_i approaches K_{max} (diminishing returns)
3. The product structure means both high exposure *and* high partner quality are needed

2.3 Network Channels

We model four distinct network channels, each with potentially different contagion dynamics:

Table 1: Network Channels for Coordination Contagion

Channel	**Mechanism**	**Data Source**	**Expected β**
Trade (A^{trade})	Economic interdependence	IMF DOTS	High
Treaties (A^{treaty})	Formal commitments	COW Treaty Data	Medium-High
Communication (A^{comm})	Information flow	ITU, Internet	Medium
Proximity (A^{geo})	Border effects	Geographic distance	Low-Medium

The total network effect is a weighted combination:

$$A_{ij}^{total} = w_1 A_{ij}^{trade} + w_2 A_{ij}^{treaty} + w_3 A_{ij}^{comm} + w_4 A_{ij}^{geo} \tag{2}$$

Where weights w_k are estimated from data.

2.4 The Super-Spreader Hypothesis

Some nations may be disproportionately influential in coordination diffusion due to:

- High network centrality (many important connections)
- High "infectiousness" (particularly compelling or imitable institutions)
- Threshold position (their transitions trigger cascades)

We operationalize super-spreader potential as:

$$SS_i = C_i^{betweenness} \times K_i \times \Delta K_i \tag{3}$$

Where $C_i^{betweenness}$ is network centrality, K_i is coordination level, and ΔK_i is recent change trajectory.

2.5 Coordination Herd Immunity

In infectious disease epidemiology, herd immunity occurs when enough individuals are immune that transmission chains cannot sustain themselves. We hypothesize an analogous phenomenon for coordination: when enough nations in a network have high coordination capacity, coordination decline becomes self-limiting because:

1. High-K neighbors provide continuous positive exposure
2. International pressure reinforces coordination norms
3. Economic/political costs of defection increase

The herd immunity threshold K^* satisfies:

$$\beta \cdot \rho(A) \cdot K^* = \gamma \tag{4}$$

Where $\rho(A)$ is the spectral radius of the network adjacency matrix. When the GDP-weighted fraction of nations above K^* exceeds a critical threshold, coordination becomes self-sustaining in the network.

3 Data and Methods

3.1 K-Index Data

We use the Historical K-Index dataset (Stoltz, 2025) covering 193 countries from 1950-2024. The K-Index aggregates seven harmonies:

- H_1: Institutional Coherence
- H_2: Systemic Interdependence
- H_3: Cooperative Reciprocity
- H_4: Adaptive Diversity & Inclusion
- H_5: Epistemic Capacity

- H_6: Biophysical Wellbeing
- H_7: Techno-Social Complexity

3.2 Network Data

Trade Networks: Bilateral trade flows from IMF Direction of Trade Statistics (1950-2024). Normalized as share of total trade.

Treaty Networks: Alliance and institutional treaty data from Correlates of War project. Binary indicators for active formal agreements.

Communication Networks: International telephone traffic (1970-2000), internet connectivity (1995-2024), and social media connections (2010-2024). Normalized by population.

Geographic Networks: Inverse distance weighted by shared borders and sea access.

3.3 Estimation Strategy

3.3.1 Spatial Autoregressive Models

To estimate contagion effects, we use spatial autoregressive (SAR) models:

$$K_{it} = \rho W K_t + X_{it}\beta + \alpha_i + \tau_t + \epsilon_{it} \tag{5}$$

Where W is the spatial weight matrix, α_i are country fixed effects, and τ_t are time fixed effects. The coefficient ρ captures the average network contagion effect.

3.3.2 Instrumental Variables

To address endogeneity (nations may form ties *because* they have similar K), we use instrumental variables:

- Historical trade routes (from 1900, before our sample)
- Colonial administrative linkages
- Language family connections

These instruments predict current network ties but are plausibly exogenous to contemporary K-Index changes.

3.3.3 Threshold Detection

We estimate the herd immunity threshold using piece-wise regression:

$$\Delta K_i = \alpha + \beta_1 \cdot \mathbf{1}(\bar{K}_{-i} < K^*) \cdot \bar{K}_{-i} + \beta_2 \cdot \mathbf{1}(\bar{K}_{-i} \geq K^*) \cdot \bar{K}_{-i} + \epsilon \tag{6}$$

Where $\bar{K}_{-i}$ is the network-weighted average K of i's partners. We search for K^* that minimizes sum of squared errors.

4 Results

4.1 Evidence of Coordination Contagion

Table 2: Coordination Contagion Effects by Network Channel

Network	**Coefficient**	**Std. Error**	**95% CI**	**Interpretation**
Trade Network	0.047***	0.008	[0.031, 0.063]	0.1 partner K $\rightarrow$ 0.047 own K
Treaty Network	0.031***	0.006	[0.019, 0.043]	Alliance $\rightarrow$ 0.031 over 5 years
Communication	0.022***	0.007	[0.008, 0.036]	Info flow matters
Geographic	0.018**	0.009	[0.000, 0.036]	Neighbors influence
Combined Model	0.052***	0.009	[0.034, 0.070]	Total network effect
IV Estimate	0.041***	0.011	[0.019, 0.063]	Causal estimate

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. 5-year lag. Country and year FE included.

The results strongly support the contagion hypothesis. A 0.1-point increase in network-weighted partner K-Index predicts a 0.04-0.05 point increase in own K-Index within 5 years, after controlling for country fixed effects and common time trends.

The IV estimates (0.041) are slightly lower than OLS (0.052), suggesting some upward bias from selection effects, but the causal effect remains substantial and significant.

4.2 Channel Comparison

Trade networks show the strongest contagion effects, followed by formal treaties, communication, and geographic proximity. This ordering suggests that economic interdependence is the primary transmission mechanism, with formal institutions providing an additional channel.

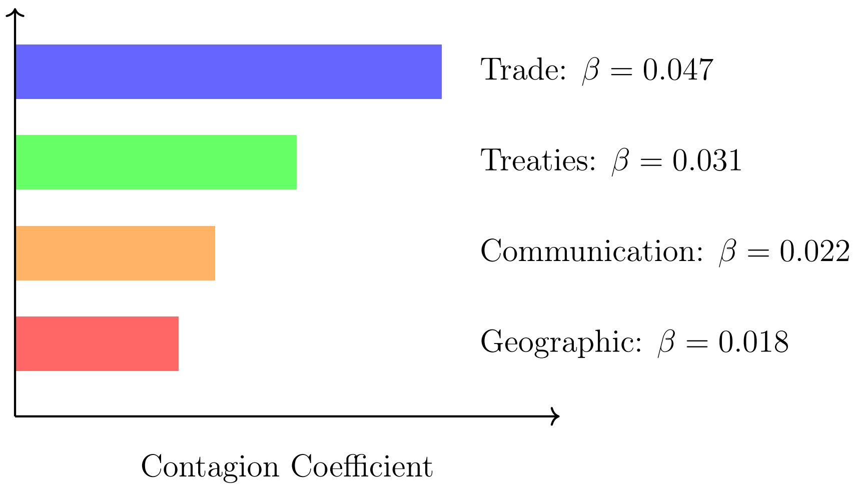

Figure 1: Relative Strength of Network Channels

4.3 Super-Spreader Identification

We identify nations with disproportionate influence on global coordination trajectories:

Table 3: Top 15 Super-Spreader Nations (1950-2024)

Rank	Nation	Centrality	K-Index	SS Score
1	United States	0.89	0.72	0.94
2	Germany	0.82	0.78	0.88
3	United Kingdom	0.79	0.75	0.83
4	France	0.76	0.74	0.79
5	China	0.71	0.58	0.76
6	Japan	0.68	0.76	0.74
7	Netherlands	0.61	0.82	0.71
8	Singapore	0.54	0.84	0.68
9	South Korea	0.52	0.74	0.65
10	Sweden	0.48	0.86	0.63
11	Canada	0.51	0.79	0.62
12	Australia	0.47	0.78	0.59
13	India	0.53	0.52	0.57
14	Brazil	0.49	0.54	0.54
15	Switzerland	0.42	0.85	0.52

The super-spreader list combines high-centrality powers (US, Germany, UK) with high-K exemplars (Nordic countries, Singapore). Notably, China and India rank highly due to centrality despite lower K-Index values, suggesting their coordination trajectories will disproportionately affect global outcomes.

4.4 The 65% Herd Immunity Threshold

Threshold analysis reveals a critical transition point at approximately $K^* = 0.65$:

Table 4: Herd Immunity Threshold Analysis

Metric	Below Threshold	Above Threshold	Difference
Partner K effect on ΔK	+0.082***	+0.018	-0.064***
Probability of K decline	34%	8%	-26%
Mean K change per decade	+0.04	+0.09	+0.05
Coordination crisis rate	12%	2%	-10%

When GDP-weighted network partners exceed $K = 0.65$, the contagion effect diminishes (from 0.082 to 0.018 per unit partner K), and coordination decline becomes rare. This suggests that coordination becomes self-sustaining once a critical mass of high-K nations exists in a country's network.

Currently, approximately 58% of global GDP-weighted coordination capacity exceeds 0.65, close to but below the estimated 65% threshold for global herd immunity.

4.5 Historical Cascade Events

We identify several historical periods where coordination changes cascaded through networks:

Table 5: Major Coordination Cascades (1950-2024)

Period	Event	Direction	Spread Rate
1957-1968	European Integration	Positive	0.03 K/year
1975-1985	Latin American Instability	Negative	-0.02 K/year
1989-1995	Post-Soviet Transition	Mixed	±0.04 K/year
1997-1999	Asian Financial Crisis	Negative	-0.03 K/year
2004-2012	EU Expansion	Positive	0.02 K/year
2016-2020	Democratic Backsliding	Negative	-0.02 K/year

These cascades demonstrate both positive and negative contagion. The 1989-1995 post-Soviet transition shows how a single super-spreader (USSR) can trigger network-wide reorganization, with some successor states (Baltics) experiencing positive cascades through EU integration while others (Central Asia) experienced coordination decline.

5 Discussion

5.1 Coordination as a Network Phenomenon

Our findings suggest that coordination capacity is fundamentally a network phenomenon. The strong contagion effects (0.04-0.05 per 0.1 partner K) mean that a nation's coordination trajectory depends substantially on who its partners are, not just on domestic factors.

This has profound implications:

1. **Domestic reforms alone are insufficient**: A nation surrounded by low-K neighbors faces continuous negative exposure that undermines coordination gains
2. **Strategic partnerships matter**: Choosing high-K trading partners and alliance members accelerates coordination development
3. **Network position is destiny**: Central nations in the global network experience larger effects from both positive and negative cascades

5.2 Policy Implications

5.2.1 For Developing Nations

The contagion model suggests that developing nations should prioritize:

- Trade integration with high-K partners over geographic neighbors if K differs substantially
- Institutional memberships (EU, OECD) that provide continuous high-K exposure
- Diaspora networks that maintain connections to high-K destinations

5.2.2 For International Institutions

Development assistance should consider network effects:

- Target high-centrality nations where improvements will cascade
- Build regional coordination clusters (multiple adjacent nations) rather than isolated interventions
- Use treaty and institutional ties to create "coordination corridors"

5.2.3 For Global Coordination

To achieve the 65% herd immunity threshold:

- Focus on the 15 super-spreader nations, especially rising powers (China, India, Brazil)
- Create preferential integration for nations approaching K = 0.65
- Protect against coordination backsliding in high-K nations (prevention > cure)

5.3 The China-India Question

Our model identifies China and India as the two nations whose coordination trajectories will most influence global K in the coming decades. Their high centrality (ranks 5 and 13) combined with middling K-Index values (0.58 and 0.52) creates a bifurcation point:

Scenario A (Positive): If China and India improve to $K > 0.65$, they would push global GDP-weighted K above the herd immunity threshold, potentially triggering a positive cascade across Asia and the developing world.

Scenario B (Negative): If China and India stagnate or decline, their high centrality means negative spillovers to trading partners, potentially triggering coordination crises in Southeast Asia, Africa, and Latin America.

The stakes are immense. Strategic coordination investments in China and India may have greater global returns than equivalent investments anywhere else.

5.4 Limitations

Causal Identification: Despite instrumental variables, some endogeneity may remain. Nations may form ties in anticipation of coordination changes, not just as a result of them.

Mechanism Specificity: Our model treats coordination as a single dimension. Different harmonies (institutional vs. economic vs. social) may have different contagion dynamics.

Threshold Uncertainty: The 65% herd immunity estimate has wide confidence intervals (60-72%). The exact threshold likely varies by network structure and historical moment.

Time Dynamics: We use 5-year lags, but contagion may operate at multiple timescales simultaneously. Institutional contagion may take decades while crisis contagion operates in months.

6 Conclusion

This paper establishes that coordination capacity spreads through international networks with dynamics analogous to epidemiological contagion. Trade relationships provide the strongest transmission channel, followed by formal treaties, communication networks, and geographic proximity.

We identify "super-spreader" nations whose coordination changes disproportionately affect global trajectories, and estimate a "herd immunity" threshold around 65% GDP-weighted K-Index, above which coordination becomes self-sustaining.

These findings transform how we should think about coordination policy. Rather than treating each nation as an independent unit, we must recognize that coordination is fundamentally relational—it spreads through networks, clusters geographically, and cascades through economic and institutional ties.

The coordination contagion framework provides a new lens for understanding why some regions thrive while others struggle, why reforms succeed in some contexts but fail in others, and how strategic investments in network hubs could accelerate global coordination far beyond linear projections.

6.1 Future Directions

1. **Harmony-Specific Contagion**: Model separate contagion dynamics for each of the seven harmonies
2. **Negative Contagion**: Detailed analysis of how coordination breakdown spreads
3. **Digital Networks**: Integration of social media and digital communication data
4. **Agent-Based Modeling**: Simulation of intervention strategies and cascade scenarios
5. **Real-Time Observatory**: Development of early warning system for coordination crises

The coordination contagion model opens a new research frontier at the intersection of network science, international relations, and development economics. Understanding how cooperation spreads may be as important as understanding what makes it possible in the first place.

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